# Skill 3: Customer Churn using Logistic Regression

A telecom company wants to predict whether a customer will leave. We use the Breast Cancer Wisconsin dataset only as a clean classroom binary-classification dataset. The same workflow can be transferred to churn data.

In [ ]:
```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,roc_auc_s
```

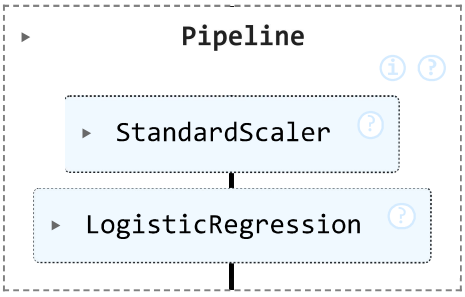
In [2]:
```
data=load_breast_cancer()
```

In [ ]:
```
X, y = data.data, data.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=42
)
```

In [5]:
```
log_reg = Pipeline([
    ("scaler", StandardScaler()),
    ("model", LogisticRegression(max_iter=2000))
])
```

In [7]:
```
log_reg.fit(X_train, y_train)
```

Out[7]:




In [8]:
```
y_pred = log_reg.predict(X_test)
y_prob = log_reg.predict_proba(X_test)[:, 1]
```

In [9]:
```
print("Accuracy :", round(accuracy_score(y_test, y_pred), 4))
print("Precision:", round(precision_score(y_test, y_pred), 4))
print("Recall :", round(recall_score(y_test, y_pred), 4))
print("F1-score :", round(f1_score(y_test, y_pred), 4))
print("ROC-AUC :", round(roc_auc_score(y_test, y_prob), 4))
```

```
Accuracy : 0.9825
Precision: 0.9861
Recall : 0.9861
F1-score : 0.9861
ROC-AUC : 0.9954
```

In [10]:
```
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| malignant | 0.98 | 0.98 | 0.98 | 42 |
| benign | 0.99 | 0.99 | 0.99 | 72 |
| accuracy | | | 0.98 | 114 |
| macro avg | 0.98 | 0.98 | 0.98 | 114 |
| weighted avg | 0.98 | 0.98 | 0.98 | 114 |